SYED MOIZ ALI

Principal AI/ML Engineer & Solutions Architect| IIT Kanpur

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SUMMARY

Principal AI/ML Engineer and Systems Architect with nearly a decade of experience designing, scaling, and deploying production-grade machine learning ecosystems and cloud infrastructures. Merges a rigorous academic foundation in Operations Research with deep expertise in Generative AI, LLMOps, and distributed systems. Proven track record of accelerating product cycles through Agentic Engineering, orchestrating cross-functional technical teams, and architecting high-availability cloud environments. Highly skilled in transitioning complex statistical models and RAG architectures into resilient, automated enterprise SaaS platforms.

TECHNICAL SKILLS

- Generative AI & LLMs: RAG Architectures, LLM Evaluation & Alignment, Prompt Engineering, Autonomous AI Agents, Transformers (Hugging Face).
- Machine Learning & Optimization: XGBoost, LightGBM, Time-Series Forecasting, CNNs, LSTMs, Genetic Algorithms, Optuna, Ray Tune, Hyperopt, SHAP, LIME.
- Cloud Architecture & MLOps: AWS (EC2, ECS, SageMaker, RDS, VPC, S3), Docker, Kubernetes, Terraform (IaC), CI/CD (GitHub Actions).
- Software & Data Engineering: Python, FastAPI, PostgreSQL, Apache Spark, Data Pipeline Orchestration.

PROFESSIONAL EXPERIENCE

Independent AI Researcher & Open-Source Contributor

***Snorkel AI, PaperBench & AI Ventures* | 05/2026 – Present**

- Advanced foundational model safety and evaluation consistency, as measured by enhanced task reproducibility across AI benchmarks, by engineering high-fidelity programmatic labeling datasets as an expert contributor for Snorkel AI.
- Strengthened evaluation environments for automated code generation, as measured by improved validation metric accuracy, by integrating robust test suites and validation modules into the PaperBench machine learning framework.
- Accelerated concept-to-deployment execution for autonomous workflows, as measured by rapid prototype launches at the Dreamflow Buildathon 2025, by architecting AI agents and integrating emergent Generative AI frameworks.

CTO & AI Systems Architect

***RegTech Innovations Pvt Ltd.* | 07/2025 – 05/2026**

- Orchestrated the end-to-end delivery of the OutsourceGuard enterprise SaaS platform, as measured by accelerated time-to-market rollout, by executing an Agentic Engineering workflow to rapidly scaffold application layers around core AI engines.
- Optimized enterprise contract processing throughput, as measured by a 40% reduction in manual document review time, by architecting Vendor Risk Modeling systems using fine-tuned Hugging Face Transformers for automated risk scoring.
- Ensured high-availability infrastructure stability, as measured by 99.9% system uptime validated via CloudWatch, by provisioning foundational cloud environments on AWS alongside custom domain routing and automated GitHub Actions CI/CD pipelines.

Senior Machine Learning Consultant

***Studypool Inc.* | 08/2019 – 06/2025**

- Improved industrial time-series forecast reliability, as measured by double-digit performance gains over baseline models, by designing and deploying complex ensemble architectures integrating gradient boosting and LSTM networks.
- Enhanced domain-specific conversational AI performance, as measured by an 18% increase in internal NLP accuracy benchmarks, by deploying custom RAG architectures and leveraging advanced prompt engineering techniques.
- Streamlined system provisioning workflows, as measured by a 60% reduction in deployment time and an 85% drop in pipeline failures, by migrating legacy environments to Terraform and deploying containerized services on AWS ECS and SageMaker.
- Elevated document classification accuracy, as measured by a 94% precision rate, by engineering scalable CNNs integrated with Optuna hyperparameter tuning and SHAP explainability frameworks.
- Scaled backend model inference capabilities, as measured by 99.2% API availability during 300% traffic spikes, by orchestrating FastAPI microservices across Docker and Kubernetes clusters.

Data Consultant
Studypool Inc. **| 06/2017 – 07/2019**

- Accelerated transactional data retrieval, as measured by a 32% reduction in average query execution times, by executing comprehensive schema refactoring and advanced B-tree indexing in PostgreSQL.
- Eradicated critical reporting discrepancies and pipeline outages, as measured by zero downstream alert triggers over consecutive quarters, by leveraging Apache Spark to orchestrate automated, high-volume data processing workflows.

Research Assistant
Sultan Qaboos University (Muscat, Oman) **| 11/2014 – 12/2015**

- Optimized dynamic production balancing and order acceptance, as measured by simulated profit margins, by formulating and executing complex mathematical optimization models leveraging statistical methods.
- Accelerated multi-scenario decision evaluations under operational constraints by programming and executing Genetic Algorithms in MATLAB.

EDUCATION

Master of Technology (M.Tech) in Operations Research (CPI: 8.0)
Indian Institute of Technology (IIT), Kanpur, India | 2009-2011

Bachelor of Engineering (B.E.) in Electronics & Telecommunication (Honors: 8.5)
CSVTU, Bhilai, India | 2005 – 2009

SELECT CERTIFICATIONS

- Machine Learning Engineering for Production (MLOps) - DeepLearning.AI
- Deep Learning Specialization - DeepLearning.ai
- TensorFlow Developer Specialization - DeepLearning.AI
- Generative Adversarial Networks (GANs) Specialization - DeepLearning.AI
- Algorithms Specialization – Stanford Online
- AI Foundations for Business Specialization – IBM

SELECT PUBLICATIONS

- Sharma, R. R. K., & Ali, S. M. (2017). *Reducing a Lot Sizing Problem with Set-up, Production, Shortage, and Inventory Costs to Lot Sizing Problem with Set-up, Production, and Inventory Costs*. American Journal of Operations Research, 7, 282-284. Link
- Ali, S. M., Sharma, R.R.K., & Gupta, O.K. (2015). *Lagrangian Relaxation Procedure for the Capacitated Dynamic Lot Sizing Problem*. AIMS International Conference on Management. Link
- Syed, M. A., & Sharif. (2012). *Aggregate planning for semi-finished goods in a make-to-stock environment.* International Journal of Advances in Management, Technology & Engineering Sciences, 1(8(I)), 104-107.
- Sharif, & Syed, M.A. (2012). Procurement Policies & Inventory Management System in Manufacturing and Service Settings: An Optimization Framework. International Journal of Business, Management & Social Sciences, 1(9), 27- 32.